Geometric Deep Learning: from (Bio)Medical Dataset Building to Downstream Applications

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Outline

- **Introduction**
  - Expectation of this deep dive
  - Importance of dataset development
- **Dataset Building Steps**
- **Example**
  - Development of *RibSeg* dataset
  - Geometric deep learning-based baseline model
  - Evaluation metrics and test results for the baseline model
  - Future works on downstream tasks
- **Summary**

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Introduction - Expectation of this Deep Dive

Readiness Assessment

- **The importance of developing new datasets**
- **A general idea on the procedure of building datasets**
  - Summarized to 5 Steps
  - E.g. Building a large scale dataset on **rib segmentation and rib centerline extraction** from scratch
- **How geometric deep learning approach and the medical dataset can facilitate each other's development**
  - E.g. Designing a geometric deep learning-based baseline model

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Introduction - Importance of Dataset Development

Readiness Assessment

- **Deep learning research is driven by large datasets**
  - Model developing and benchmarking
- **Explore new research questions**
  - Create new datasets or extend existing datasets
- **Benchmarking existing methods**
  - Model comparison
- **Understanding the pitfalls of existing materials and methods**
  - Are existing methods/datasets good enough for real-world applications?
  - (e.g. subtle details, domain generalization, model calibration,...)
- **Commercial use / building academic influence**
  - Datasets and benchmarks are generally highly-cited (e.g. Imagenet)

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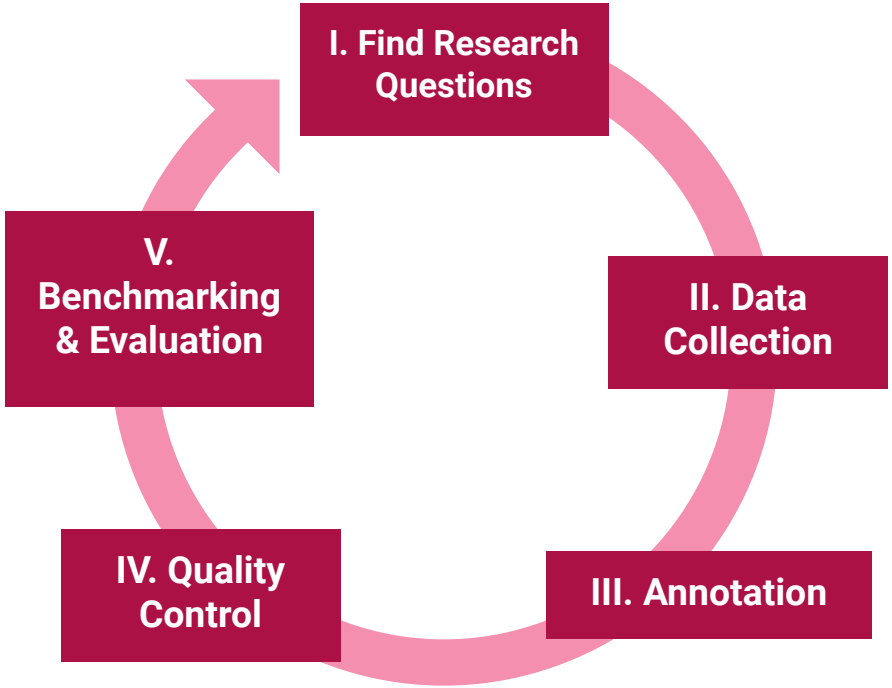
Dataset Building Steps

5 steps to build a dataset

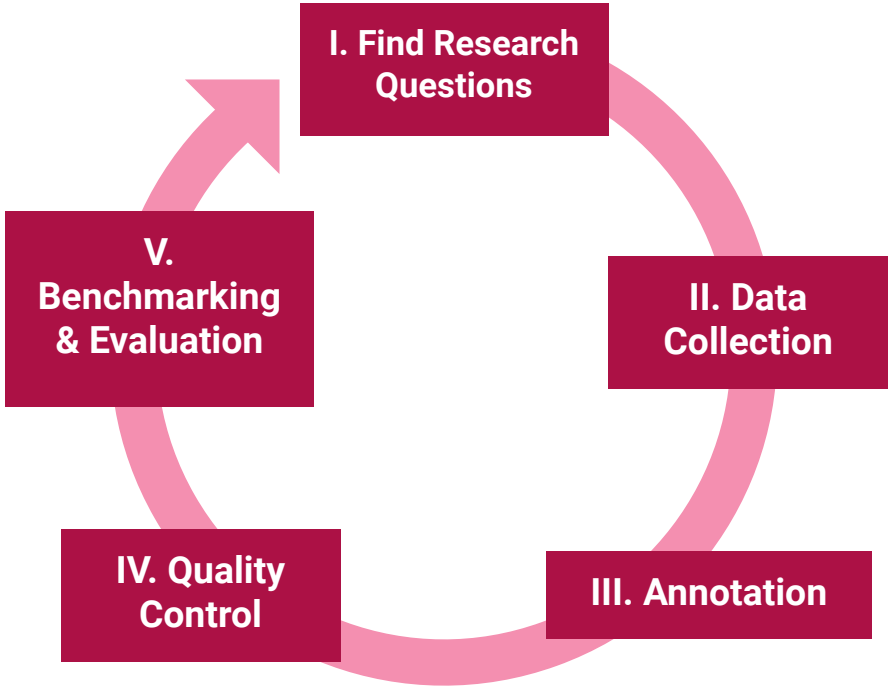
- **I. Find Research Questions**
  - Existing dataset? – Extend or Create
  - Is the problem important and solvable?
- **II. Data Collection**
  - Public or in-house data – Privacy / ethics
  - Inclusion criteria – real-world distribution
  - Single-center or multi-center – Heterogeneity
  - How large?
    - ***Rule of thumb in clinical AI:***
    - ***At least 300 patients***
    - ***Medical Center Scale: > 100,000***
- **III. Annotation**
  - Who should be the annotators?
    - Experts, anyone after training
  - How many annotators?
    - ***Rule of thumb in clinical AI:***
    - ***At least 1 junior + 1 senior (to confirm)***
  - How to make the annotation procedure fast and scalable?
  - Is the annotation already existing in certain ways?



5 steps to build a dataset

- **IV. Quality Control**
  - How to keep the annotation consistent?
    - ***Facts in clinical AI:***
    - **Highly ambiguous**
    - **Unstable annotation quality**
  - **Human-in-the-loop** quality control
  - **Model-based** quality control

- **V. Benchmarking & Evaluation**
  - Design **metrics** – real-world scenario
  - Introduce **baseline methods** – state of current algorithms?
  - Benchmark – **standardized** for future research?



Example - Development of *RibSeg* Dataset I

I. Find Research Questions

Background:

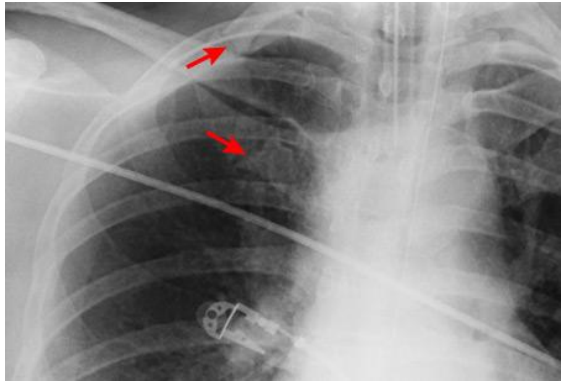
Manual rib inspection in CT scans is **labor-intensive**

Challenges:

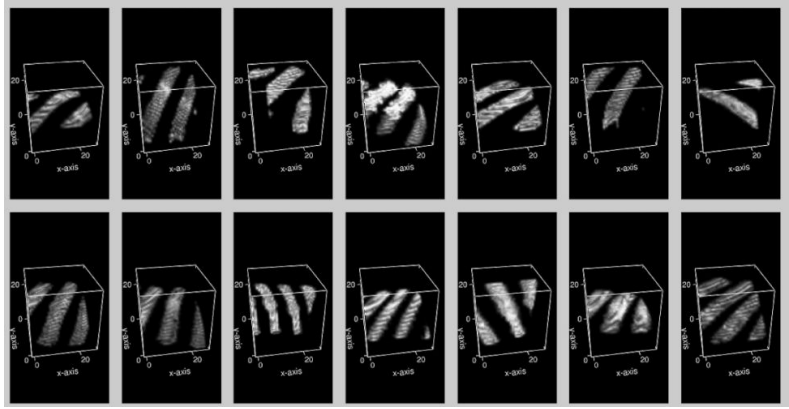
1. **Lack of labeled dataset** for developing automatic rib segmentation
2. Previous segmentation methods work on dense 3D volumes that are **computationally inefficient**

Ideas:

1. Develop a labeled large **rib segmentation benchmark**
2. Designed a **point cloud-based baseline method** for rib segmentation



Rib Inspection in CT Scans for Fracture Detection[1]



Sliding Window Visualization on Ribs

[1] Illustration by Peter Cundy, MBBS, FRACS: https://bestpractice.bmj.com/topics/en-us/1009

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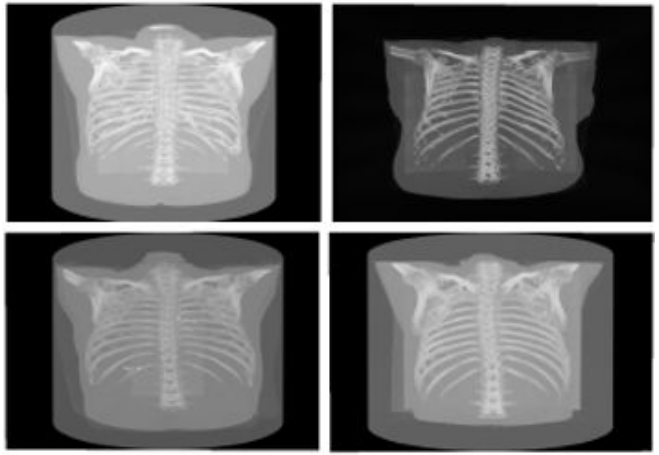
Example - Development of *RibSeg* Dataset II

Readiness Assessment

II. Data Collection

- **CT Scans from *RibFrac* dataset, previous work for rib fracture detection**
- **Contain many traumatic rib fractures**
- **Sufficient chest-abdomen CT scans for dataset development**



RibFrac Dataset[2]

Cohorts	Availability	No. Patients / CT Scans	No. CT Slices	No. Fractures
Training	Total	720	265,302	6,156
	Public	420	154,127	3,987
	In-House	300	111,175	2,169
Tuning	Public	60	22,562	435
Test	Public	120	44,619	882

RibFrac Dataset Distribution[2]

[2] Jin L, Yang J, Kuang K, et al. Deep-learning-assisted detection and segmentation of rib fractures from CT scans: Development and validation of FracNet. EBioMedicine (by Lancet)

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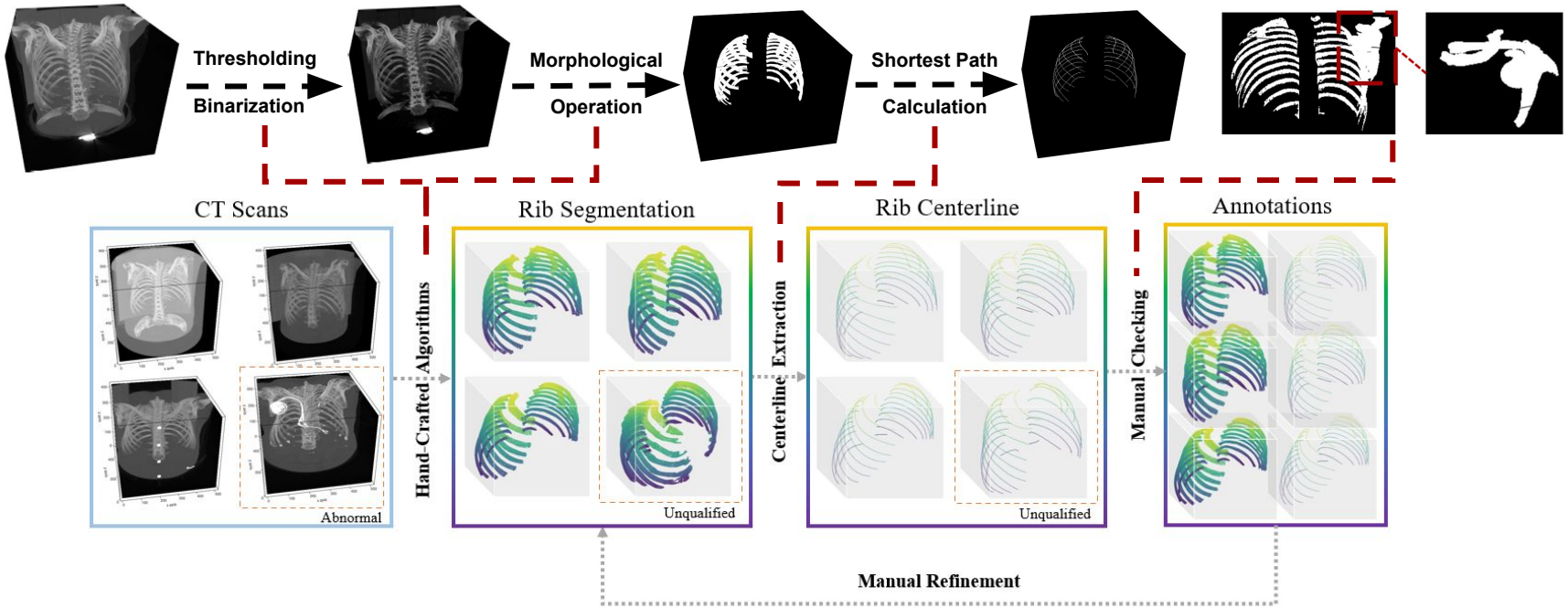
III. Annotation & IV. Quality Control

- **Pipeline of *RibSeg* Development**



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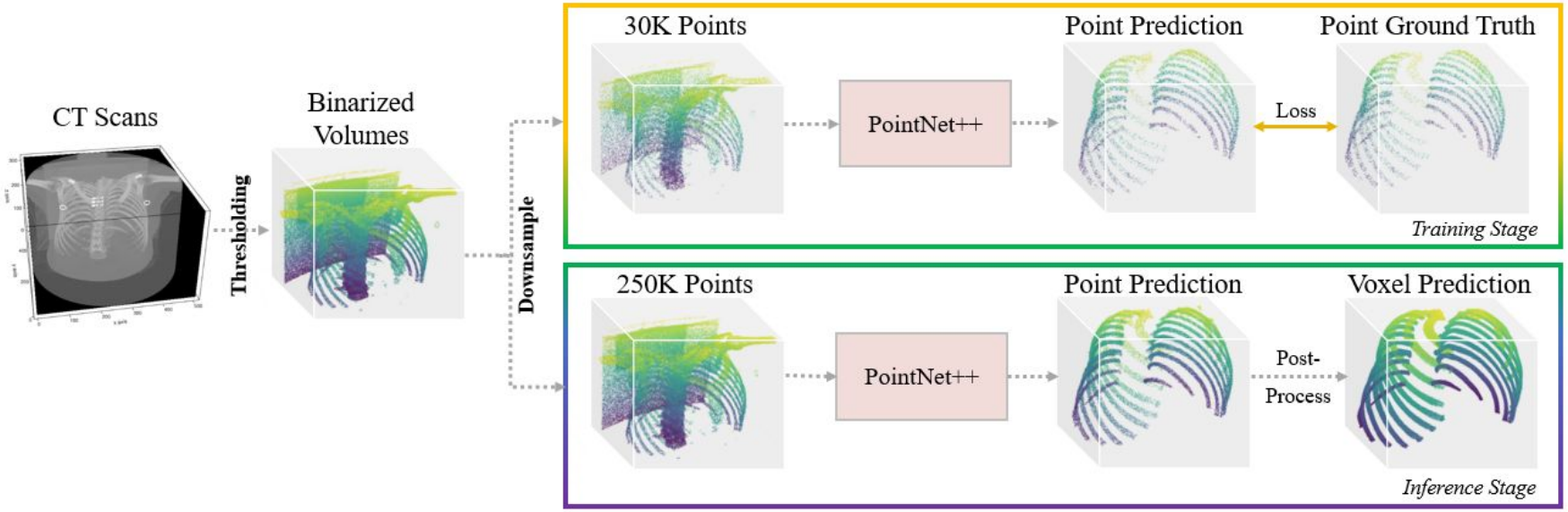
V. Benchmark & Evaluation

- **Pipeline of Baseline Model Development**



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Example - Evaluation Metrics and Results

Readiness Assessment

V. Benchmark & Evaluation

- **Metrics for rib segmentation**
  - Dice over sparse points (Dice(P)), and dense voxels (Dice(v))
  - Ratio of missed rib pairs at recall> 0.5
  - The model forward time in second
- **Results of baseline models**

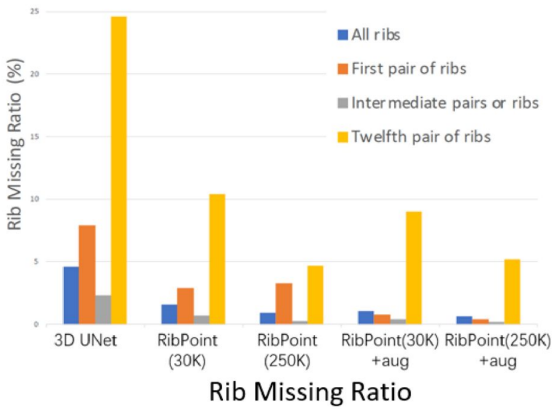
Dice Coefficient

$$\text{Dice}^{(L)} = 2 \cdot \left|y_{(L)} \cdot \hat{y}_{(L)}\right| / \left(\left|y_{(L)}\right| + \left|\hat{y}_{(L)}\right|\right), L \in \{P, V\},$$

where $Dice^{(P,V)}$, $\hat{y}_{(P,L)}$ *and* $y_{(P,L)}$ indicate point-wise and voxel-wise Dice coefficient, prediction, and ground truth, respectively.

Rib Missing Ratio

A missing of an individual rib i is counted if $recall_i < 0.5$.
The missing ratio of all / first / intermediate / twelfth pairs of rib is calculated and reported.



Methods	$Dice^{(P)}(std)$	$Dice^{(V)}(std)$	Missed Ribs (A/F/I/T)	Forward (s)
Morphology-Based	-	-	-	> 500
Voxel-Based 3D UNet[10, 23]	-	86.2 (0.11)%	4.6%/7.9%/2.3%/24.6%	30.63
RibPoint (30K)	92.3 (0.04)%	91.0 (0.03)%	1.6%/2.9%/0.7%/10.4%	**0.32**
RibPoint (250K)	91.6 (0.08)%	92.4 (0.09)%	0.9%/3.3%/0.3%/4.7%	1.12
RibPoint (30K) + aug.	**94.8 (0.05)%**	93.2 (0.03)%	1.1%/0.8%/0.4%/9.0%	**0.32**
RibPoint (250K) + aug.	94.6 (0.04)%	**95.2 (0.02)%**	**0.6%/0.4%/0.2%/5.2%**	1.12

Quantitative Result

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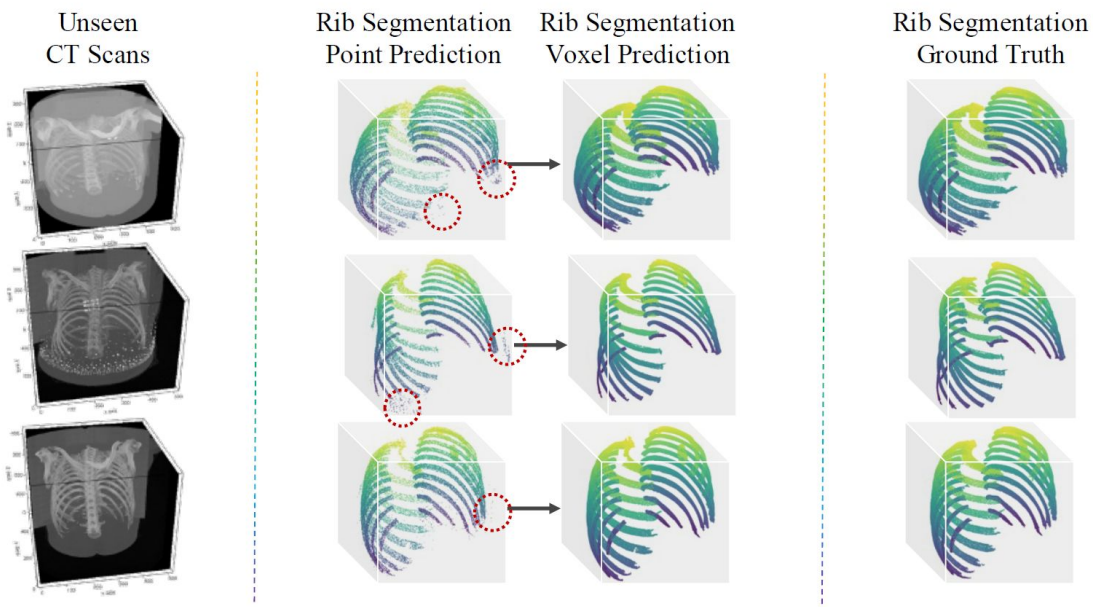
Example - Downstream works

Readiness Assessment

- **Use the trained model to generate segmentations which are difficult to manually annotate**
- **Develop centerline extraction model**
- **Backbone modification for better accuracy**
- **Extend dataset with multi-center data and improve the model robustness by domain adaptation methods**



Generate segmentations by trained model

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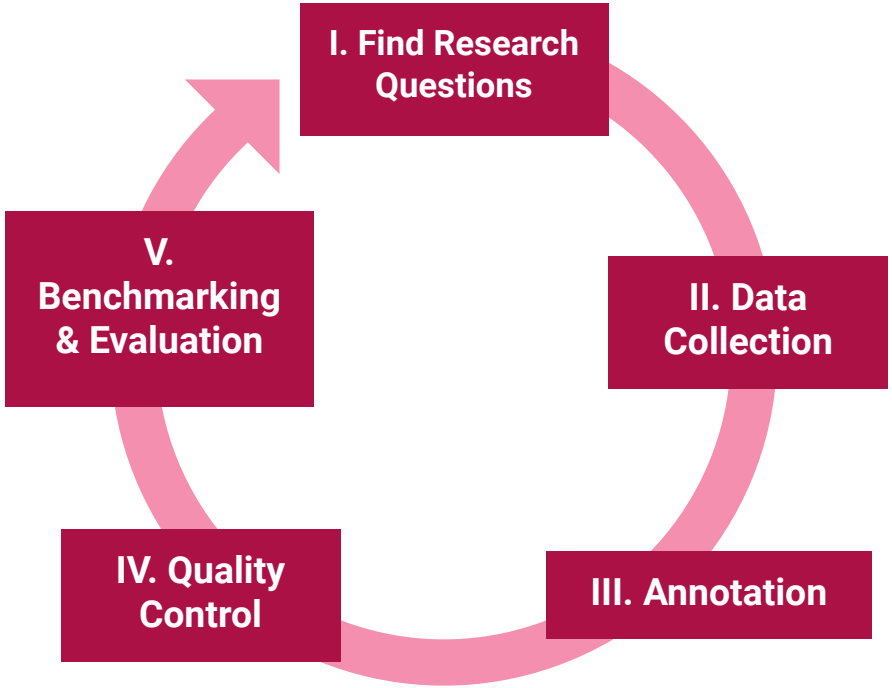
Summary

- The importance of developing new datasets
- The procedure of building datasets
- How geometric deep learning approach and the medical dataset can facilitate each other's development



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Thanks for listening!
Happy new semester!

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